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Climate is not a straight line: Scalable Python-based GAMM Workflows for Wildlife Conservation

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Abstract

Wildlife conservation organizations are collecting growing volumes of ecological data, but technical barriers limit their effective use for decision-making. We present a Python-based trend fitting framework designed for non-technical conservationists that operationalizes Generalized Additive Mixed Models (GAMMs) within a scalable and reproducible pipeline. The framework automates model fitting, evaluation, and hyperparameter selection, and is fine-tuned for conservation use cases. Through a primary case study on nine Kenyan forest cover sites, and a secondary robustness check on elephant movement telemetry, we demonstrate how GAMMs improve data fitting compared to other common statistical baselines. We conclude by discussing how we implemented our framework within the Ecoscope library, which enabled the large-scale deployment of our analysis pipeline worldwide.

Keywords Generalized Additive Mixed Models, Ecological Trend Analysis, Wildlife Conservation, Time Series, Bayesian Optimization

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1. Introduction

1.1. Background

Environmental research organizations are recording an increasing amount of data on animal movements and landscape changes. These real-world monitoring systems generate sparse and irregular data across species, terrain, and sensors. Statistical tools, such as trend fitting analysis, can leverage these datasets to help conservationists better anticipate how ecosystem dynamics may evolve over time. Their widespread adoption, however, remains constrained by technical barriers. Access to qualified personnel capable of analyzing these datasets and extracting robust conclusions remains limited. This challenge highlights an urgent need for analysis tools capable of transforming noisy and complex datasets into insights without requiring the user to have advanced expertise in statistics and programming.

We present a Python-based trend fitting framework targeted at non-technical wildlife conservation practitioners. Building on the SciPy [1], Scikit-Learn [2], [3], and StatsModels [4] libraries, our pipeline modularizes model fitting, hyperparameter selection, and evaluation. We then present interpretable results through an interactive and reproducible dashboard. At its core, our framework utilizes Generalized Additive Mixed Models (GAMMs), a powerful statistical approach for modeling complex, random effect relationships. We introduce a robust GAM/GAMM Regressor class that automates smoothing parameter selection over large matrices of data. Through a primary case study on forest cover across nine monitoring sites, we show how GAMMs can be a scalable and interpretable improvement on other common time series models. This is especially applicable in noisy, low-data, and multi-entity regimes often encountered in conservation databases. A secondary case study on elephant

movement speed, tracking over the course of one year, demonstrates the robustness of this pipeline by applying it unmodified to a dataset that differs from the forest series in sampling frequency, response variable, and noise structure. We show the relative improvements GAMs offer in comparison to Generalized Linear Models (GLMs) and standard linear regression models (OLS), as well as the challenges of considering deep learning regression analysis for ecological data.

We further discuss how our design decisions were shaped by trade offs between interpretability, robustness, and predictive performance. This includes topics like mitigating the risks of overfitting and maintaining user trust.

Finally, we explain how we implemented our framework within Ecoscope, an open-source Python library for conservation analytics. We provide a brief overview of the package architecture and demonstrate how Ecoscope enables the successful real-world deployment of our trend fitting pipeline within the EarthRanger monitoring platform, which is currently being used by more than 800 wildlife conservation sites across the globe.

2. Related Work

2.1. EarthRanger and the Ecoscope Library

EarthRanger is an open-source software platform developed by the Allen Institute for AI and Wildlife Dynamics. It is used to support protected area managers, ecologists, and wildlife biologists in analysis. The system consists of seven main components — Core Server, API, Storage, Gundi, Web App, Mobile App, and Ecoscope — providing functionality for data handling and storage, real-time and post-collection analysis, visualization, and sharing. EarthRanger handles a variety of data including GPS telemetry, ranger patrol data, camera traps, and remote sensing.

Ecoscope is the open-source Python analytics layer of EarthRanger. Its modules are used for analysis workflows related to wildlife movement and conservation datasets. It can handle data from major Earth reporting data lakes including EarthRanger, Google Earth Engine, MoveBank, and GeoPandas. The library supports conservationists as a visualization tool with established workflows unique to each study, and has been integral in reducing analysis time in a wide range of research and conservation fields.

The forest cover data used in this study is derived from the University of Maryland Hansen Global Forest Change dataset, accessed via Google Earth Engine. This dataset provides annual, 30-meter resolution estimates of forest cover, loss, and gain globally from 2000 onward, enabling consistent longitudinal analysis across geographically dispersed conservation sites [5]. The elephant walking speed dataset is provided by Wildlife Dynamics and field researchers from the Mara Elephant Project, and consists of hourly GPS relocations for a single collared bull elephant recorded between March 2008 and April 2009.

2.2. Trend Analysis for Ecological Datasets

Long term trend estimation in conservation is a fundamental challenge due to data sparseness, irregularity, and comparability. Measurement noise is common due to tool errors and interaction from other abiotic and biotic factors. Additionally, there exists a massive obstacle in communicating findings to sites exhibiting similar traits.

In choosing a well fitting and robust regression model we consider the relative strengths and increased complexity of four curves: standard linear regression (OLS), Generalized Linear Models (GLM), Generalized Additive Models (GAM), and Generalized Additive Mixed Models (GAMMs).

Standard linear regression, while widely used, assumes a constant rate of change over time and a Gaussian error structure. These assumptions are routinely violated in ecological datasets exhibiting nonlinear dynamics and regime changes, such as natural disasters, climate changes, or an influx of human influence.

Generalized Linear Models (GLMs) extend the linear framework to accommodate non-Gaussian response distributions by using a link function and exponential family distribution [6]. GLMs have been applied across ecology for modeling count data (Poisson family), presence-absence data (binomial family), and positive continuous measurements (gamma family). However, the linearity constraint on the link function again limits its ability to represent trends that do not follow a linear relationship.

2.3. Modeling Non-linear Dynamics

Generalized Additive Models (GAMs) address the linearity limitation of GLMs by replacing the parametric linear predictor with a sum of smooth nonparametric functions estimated from data [7], [8]. GAMs have seen increasing application in ecology for modeling species distribution responses to environmental and phenological trends. Since they produce smoother curves with quantified uncertainty, they are particularly useful for deployment in field researcher use. It is important to note that all three discussed models are site specific.

The extension to Generalized Additive Mixed Models (GAMMs) introduces random effects to the GAM framework, separating a smooth trend shared across all sites from site-specific deviations. This allows for multiple correlated observational units — sites, subjects, or repeated measurements — to improve statistical strength. Forrest et al. [9] demonstrated that GAMMs substantially reduce bias in trend estimation compared to fixed effects models in a biomedical context. This motivates their application to ecological time series with analogous properties. Baayen and Linke [10] provide a detailed outline of GAMM fitting in the context of linguistics, highlighting the role of random smooth terms in capturing deviations from population level trends.

Recent work has also explored the intersection of additive models and modern machine learning. Knieper et al. [11] proposed gradient boosting for generalized additive mixed models, showing improved predictive performance in iterative sites. Agarwal et al. [12] introduced Neural Additive Models, which replace spline basis functions with neural network components while preserving the additive interpretability structure of GAMs. Xiong et al. [13] extended mixed effects modeling to neural networks, demonstrating that random effect structures can be incorporated into deep learning architectures. These approaches represent natural future extensions of the framework presented here, though their data requirements exceed what is currently available at these EarthRanger sites.

2.4. Model Progression

Each model in this progression relaxes exactly one assumption from its predecessor, enabling progressively more flexible representations of ecological time series data.

Ordinary Least Squares (OLS): The standard linear regression model assumes a Gaussian response with an identity link and a strictly linear relationship between the predictor and the conditional mean, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, with parameters estimated by minimizing the residual sum of squares.

Generalized Linear Model (GLM): The GLM extends OLS by introducing a link function $g(\cdot)$ relating a linear predictor to the conditional mean μ_i, and by allowing the response to follow any distribution in the exponential family. In this work we use a Gaussian family with a log link, giving $\log(\mu_i) = \beta_0 + \beta_1 x_i$.

Generalized Additive Model (GAM): The GAM replaces the linear predictor with a smooth nonparametric function $f(x_i)$, so that $g(\mu_i) = \beta_0 + f(x_i)$, where f is represented as a weighted sum of K B-spline basis functions. To prevent overfitting, a smoothing penalty scaled by parameter $\alpha \geq 0$ is applied.

Generalized Additive Mixed Model (GAMM): The GAMM extends the GAM to multi-site data by decomposing the smooth trend into a global component shared across all sites and site-specific random intercepts:

$$g(\mu_{ij}) = \beta_0 + f(x_{ij}) + b_j, \qquad b_j \sim \mathcal{N}(0, \sigma_b^2) \tag{1}$$

where i indexes observations, j indexes sites, $f(x_{ij})$ is the population-level smooth, and b_j is a site-specific random intercept shrunk toward zero.

3. Regression Class Design for Ecoscope

The trend fitting module in Ecoscope is built on top of a set of scikit-learn compatible regressor classes [2], [3]. Each of the four models follows the standard `fit` and `predict` interface, which includes metrics for estimating model fit strength. The four classes — `LinearRegressionRegressor`, `GLMRegressor`, `GAMRegressor`, and `GAMMRegressor` — share common characteristics in handling preprocessing, normalization, and hyperparameterization internally. Conservationists interact only with the raw data inputs and interpretable outputs using the workflow GUI.

This section describes the key design decisions behind these classes, evaluating the tradeoffs between model flexibility and stability. Primary design decisions include normalization, smoothing parameter selection, and evaluation techniques.

3.1. Alpha Selection and Normalization

The GAM smoothing parameter α controls the balance between curve flexibility and smoothness. Values near 10^{-6} produce maximally flexible curves that closely follow every fluctuation in the data (also described as “wiggly”). Values near 10^{4} produce nearly linear curves. Selecting an appropriate α is critical: too low causes overfitting, too high produces underfitting that misses nonlinear dynamics.

Prior to fitting, all inputs are normalized within each regressor class. The time variable X is shifted to start at zero by subtracting its minimum value, and for Gaussian-family fits the response variable y is standardized to mean zero and unit variance. Shifting the time axis is required because knot boundaries are computed in shifted coordinates, and it additionally improves the conditioning of the spline design matrix. The same transformations apply to the elephant dataset, where the time axis is derived from raw timestamps stored as nanoseconds since the Unix epoch and converted to days since the first GPS fix, so the series begins at day zero by construction.

Normalization is applied internally and automatically inverted during prediction, so users always interact with inputs and outputs in their original units.

Once axes are standardized, the optimal α is selected via grid search over 100 candidate values drawn from a logarithmic grid spanning $[10^{-6}, 10^{4}]$. For each candidate value, a GAM is fit and scored using the evaluation metric specified by the user. The candidate producing the best score is selected and the final model is refitted on the full training data using that value.

For datasets with ten or fewer observations, leave-one-out cross-validation [14] is used to maximize the use of available data. For larger datasets, time-series-aware cross-validation [2], [14] via scikit-learn’s `TimeSeriesSplit` is used with five splits. This ensures that test obser-

vations always occur after their corresponding training observations and ensures realistic forecasting performance.

During rolling origin cross-validation, the GAM is trained on a temporal subset of the data but predicts observations within the full time range. α is reselected via grid search on each training window rather than fixed from a full-series fit. To prevent out of bounds errors in spline evaluation, the lower and upper knot boundaries are set to the full dataset range rather than the training subset range. Boundaries passed as raw year values are converted to the normalized scale before being passed to the spline constructor. A sliding window trains the model on N years and predicts values for the next 3 years.

3.2. Evaluation Measures

Four evaluation measures are implemented across all regressor classes.

Akaike Information Criterion (AIC) rewards model fit while penalizing wiggliness. Lower AIC indicates a better model given its complexity. AIC is only meaningful when comparing models fit on the same dataset and should not be used to compare models across sites.

$$AIC = -2\ell\left(\hat{\beta}\right) + 2p \tag{2}$$

where $\ell\left(\hat{\beta}\right)$ is the maximized log-likelihood and p is the number of effective parameters.

Bayesian Information Criterion (BIC) applies a more severe penalty for model complexity than AIC, scaled by the log of the sample size. Like AIC, it is only valid for within-dataset comparisons.

$$BIC = -2\ell\left(\hat{\beta}\right) + p\log(n) \tag{3}$$

Coefficient of Determination (R^2) measures the proportion of variance in the response. $R^2 = 1.0$ indicates a perfect fit; $R^2 = 0.0$ indicates the model performs no better than predicting the mean. Since R^2 is normalized, it can be used across sites.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \tag{4}$$

where SS_{res} is the sum of squared prediction errors and SS_{tot} is the sum of squared deviations from the mean.

Mean Squared Error (MSE) measures the average squared difference between predicted and actual values. MSE is used as the scoring metric for α selection during cross-validation.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{5}$$

Note that in the rolling origin evaluation context, MAE (mean absolute error) is additionally reported due to its frequent usage by field rangers.

3.3. Rolling Origin Evaluation

Standard cross-validation strategies such as k-fold are inappropriate for time series data because they allow test observations to occur before their corresponding training observations [15]. Rolling origin evaluation addresses this by enforcing chronological ordering throughout the evaluation pipeline.

A minimum training size is specified (15 years for the forest cover work; not applied to the single-year elephant dataset). At each step, the model is trained on all observations up to a cutoff year and evaluated on the subsequent N years of observations (3 years in this work). The cutoff is then advanced by one year and the process repeats until the end of the series is reached. For each evaluation window, MAE and RMSE are computed between predicted

and observed values in the test horizon. These are averaged across all windows to produce a single score per site per model. Spline knot boundaries are held fixed to the full observation window at every origin so that the basis is identical across folds.

For the GAMM, rolling origin evaluation is conducted across all sites simultaneously. At each cutoff, the model is trained on the first N years of data for every site jointly, and evaluated on the subsequent three years for each site. This preserves the multi-site structure of the GAMM while still evaluating out-of-sample forecasting performance per site. The model always has access to all sites during training.

3.4. Robust Data Handling

This framework is designed to handle datasets encountered across EarthRanger monitoring sites globally and relies on built-in Ecoscope functionality for I/O routing and plotting.

The trend analysis module enforces guards for a variety of data reporting issues. First, not all sites have the same number of observations. The cross-validator selection described in Alpha Selection and Normalization adapts automatically to sample size with either leave-one-out or time-series-aware validation. A minimum observation warning is raised when any site has fewer than ten data points. Second, time intervals can be inconsistent or missing in reports. The framework is not restricted to annual data: the time variable X is treated as a continuous numeric input and normalized to start at zero. Missing years are handled gracefully because the spline basis is defined over the full observed range and predictions can be made at any point within that range.

For the GAMM, all sites are combined into a single DataFrame with a site identifier column before fitting. The random intercept term requires that site identities be consistent between training and prediction. For forecasts at unseen sites, the population level smooth is returned as a fallback with all site random effects set to zero.

4. Workflow Dashboard Generation

4.1. Forest Cover

We apply the trend fitting pipeline to nine Kenyan forest monitoring sites spanning 2001–2024. Forest survival area in acres is extracted from the Hansen Global Forest Change dataset [5] via Google Earth Engine, retaining only pixels whose year-2000 tree cover exceeds 60%. Annual loss within each site is accumulated and subtracted from the baseline forested area to give a survival curve. OLS, GLM, and GAM are fit independently per site. The GAMM is fit jointly across all nine sites as a shared cubic B-spline trend with a per-site random intercept, estimated by Markov chain Monte Carlo through Bambi [16] and PyMC [17].

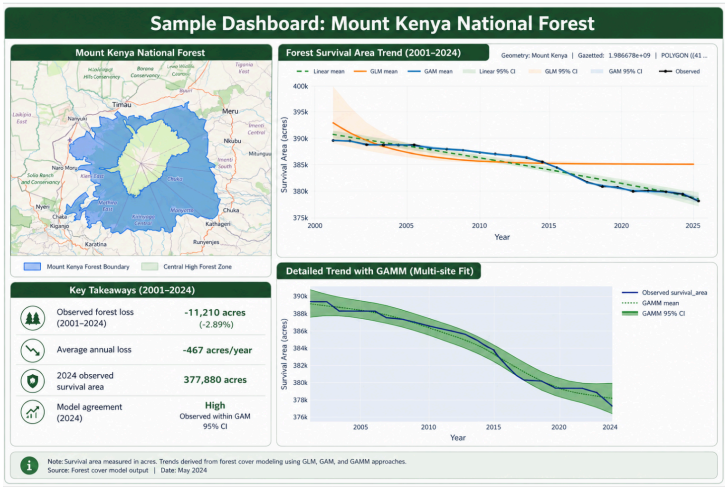


Figure 1. *Sample forest cover workflow dashboard generated from the publicly accessible workflow. This is the mock used for the Mount Kenya site for planning. It includes a map of the region, a plot displaying the three site-specific regression curves, and the multi-site GAMM fit. All show a general decline in forest cover with a relatively large drop-off between 2016–2018 and close alignment with the GAM curve.*

4.2. Elephant Walking Speed

To test whether the framework transfers beyond annual remote sensing data, we apply the identical pipeline to GPS telemetry from a single collared elephant monitored by the Mara Elephant Project between March 2008 and April 2009. The record contains 8,355 hourly movement segments, which were summarized by a daily mean travel speed to give 367 observations.

The contrast between model classes is more significant than in the forest cover case. OLS fits a nearly flat line at 0.55 km h^{-1} whose slope is statistically indistinguishable from zero. The GAM, at a selected α of 1.45, instead recovers variation with $R^2 = 0.40$.

Refitting on the unaggregated hourly segments (Figure 2, panel b) displays a similar smooth despite a doubling of observation level standard deviation (0.66 against 0.30 km h^{-1} for daily means) and an order of magnitude more observations. The stability of the recovered trend across aggregation levels indicates that the α search responds to signal rather than to sampling resolution, which supports the framework's applicability to the irregular, high-frequency telemetry typical of EarthRanger deployments. The purpose of this case study is to show that the pipeline behaves correctly on data structurally unlike the forest series, not to establish an ecological result.

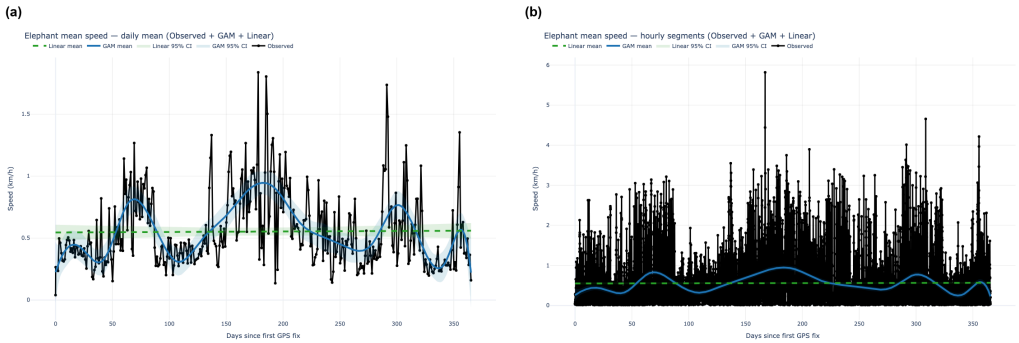


Figure 2. *Fitted OLS and GAM curves with 95% confidence intervals for elephant travel speed over one year. (a) Daily mean speed, 367 observations. (b) The same models fitted to the 8,355 unaggregated hourly*

movement segments. The linear fit is flat in both panels, while the GAM recovers comparable multi-modal structure at both aggregation levels.

4.3. In-Sample Fit

To understand how well each model type fits the nine datasets and how effective the smoothing parameter grid search is for the `GAMRegressor` class, we ran the deployed trend analysis module on each site. Table 1 reports in-sample fit metrics for the three single site models.

GAM achieves the lowest AIC on every site, confirming that flexible smooth curves provide better fit than linear alternatives even after penalizing for complexity. GAM R^2 exceeds 0.98 on all sites. OLS R^2 ranges from 0.80 to 0.98, and GLM R^2 from 0.81 to 0.99.

Selected α values are ecologically interpretable. Eburu receives $\alpha = 0.0014$ reflecting its sharp single-year drop. Samburu receives $\alpha = 95.5$ consistent with its near-flat trend. MaraConservancies receives $\alpha = 0.035$ capturing its sharp 2012–2014 cliff, and Marmanet receives $\alpha = 14.85$ despite a visible regime change. This shows how rolling origin cross-validation favors smoother curves.

Table 1. *In-sample fit metrics comparing the smoothing parameter, AIC, and R^2 for each site.*

Site	GAM α	GAM AIC	GAM R^2	GLM R^2	OLS AIC	OLS R^2
EasternMau	0.226	-48.7	0.997	0.961	3.8	0.942
Eburu	0.0014	-51.9	0.999	0.812	31.7	0.805
Loita	0.112	-27.9	0.994	0.844	26.3	0.846
MaraConservancies	0.035	-91.3	1.000	0.863	15.2	0.907
Marmanet	14.850	-21.4	0.984	0.935	16.1	0.903
MountKenya	0.722	-62.4	0.998	0.946	1.4	0.947
MountLondiani	0.572	-66.3	0.998	0.932	2.0	0.946
Narok	47.508	-53.4	0.995	0.992	-28.7	0.985
Samburu	95.455	-21.5	0.982	0.952	-1.1	0.953

4.4. Site-Level Analysis

The nine sites span a wide range of trend shapes, and the sites with abrupt transitions are where the model classes diverge most. Eburu declines sharply around 2010 and then stabilizes, so the linear models continue a decline through a period when loss had effectively stopped. MaraConservancies is the most extreme case, losing 22% of its remaining cover in the two years from 2012 to 2014. A straight line cannot represent the stable phases on either side of that cliff. Loita and Marmanet show a similar pattern, with rate changes in 2018 and 2011 respectively that the linear models absorb into a single averaged slope. At the opposite end, Narok declines almost linearly and Samburu loses less than 1% of coverage over 24 years, and these are precisely the sites where the GAM's advantage narrows and its selected α rises to 47.5 and 95.5 to produce a nearly straight fit.

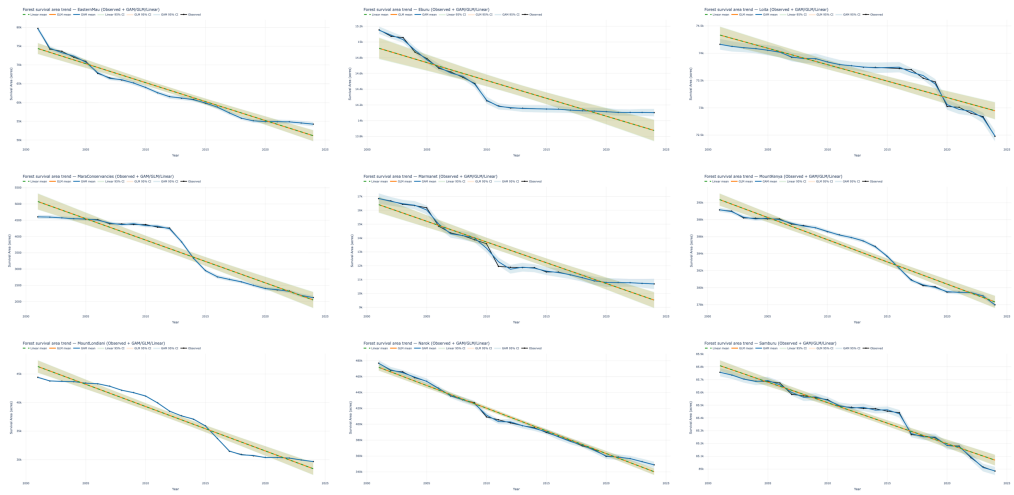


Figure 3. *Fitted OLS, GLM, and GAM curves with 95% confidence intervals for each site. Generally declining forest cover is observed across all measured sites. There are slight deviations from the linear pattern in each, with more significant drops at Eburu 2010, Loita 2018, MaraConservancies 2013, and Marmanet 2011.*

4.5. Forecasting Performance

Table 2 reports rolling origin forecasting metrics. GAM or GAMM achieves lower MAE than OLS on seven of nine sites. The two exceptions are notable: at Samburu, OLS and GAMM are essentially equivalent (92 vs 94 acres) given the near-linear trend; at MaraConservancies, OLS achieves lower MAE (304 vs 406) because after the 2012–2014 cliff the post-break trend is approximately linear.

The GLM falls between OLS and GAM on most sites. It improves on OLS where decline accelerates over time, as at Marmanet and Narok, but cannot represent a discrete break: at Eburu its MAE is 312 acres versus GAM 13. The proportional decline assumption fits gradual change but not abrupt dynamics.

The GAMM outperforms the per-site GAM on five of nine sites. Sites where GAM is superior — Eburu, EasternMau, MountLondiani, and Narok — share highly individualized dynamics not well represented by the global smooth trend. The GAMM's pooling slightly constrains these sites relative to a fully independent fit, while benefiting the remaining five through information sharing.

Table 2. *Rolling origin forecasting MAE (acres) for each of the nine forest cover sites.*

Site	GAMM MAE	GAM MAE	GLM MAE	OLS MAE
EasternMau	1534	1528	1713	2463
Eburu	30	13	312	320
Loita	180	323	342	341
MaraConservancies	379	406	502	304
Marmanet	213	441	858	1257
MountKenya	1422	1701	1476	1455
MountLondiani	2032	1834	2117	1922
Narok	5147	3730	4059	7478
Samburu	94	170	92	92

5. LLM-Based Hyperparameterization Experiment

With the increasing computing resources allocated to training large language models (LLMs), there was a question of how they may serve in hyperparameterizing the aforementioned regression classes. This section presents a preliminary empirical investigation into whether an LLM can reduce the computational cost of GAM smoothing parameter selection by proposing a narrowed α search range informed by ecological site characteristics.

The current pipeline selects α via a grid search of over 100 candidate values. This can be computationally expensive at scale when fitting 100 GAMs per site across 800+ EarthRanger deployments. The research question: can an LLM propose a credible and narrower search range (threshold of 20 values) when fed site characteristics? For each of the nine sites, a prompt was constructed describing quantitative characteristics computed from the data: site name, year range, total forest loss, mean annual loss, variance of annual loss, and the presence and timing of a regime change. Claude Sonnet 4 was selected as the model of choice due to its widespread use in industry [18]. Rounds of full grid search, LLM-guided search, and a random control group of 20 α values in a random range of equal log-width to the LLM suggestion were used to generate a comparison table.

The LLM's proposed range contained the optimal α in 2 of 9 sites, compared to 3 of 9 for the random control group. The LLM suggested $\alpha = 0.01$ for 7 of 9 sites regardless of the site-specific characteristics. It did correctly identify low-α sites such as Eburu and MaraConservancies, where regime changes are observed. The LLM failed on high-α sites such as Marmanet ($\alpha = 14.85$), Narok ($\alpha = 47.5$), and Samburu ($\alpha = 95.5$).

Table 3. *Optimal alpha values and LLM prediction accuracy for the GAM smoothing parameter.*

Site	Optimal α	LLM Guess	LLM Range	LLM Hit	Random Hit
EasternMau	0.226	0.001	[0.0001, 0.01]	N	N
Eburu	0.0014	0.001	[0.0001, 0.01]	Y	N
Loita	0.112	0.010	[0.001, 0.1]	N	N
MaraConservancies	0.035	0.010	[0.001, 0.1]	Y	Y
Marmanet	14.850	0.010	[0.001, 0.1]	N	N
MountKenya	0.722	0.010	[0.001, 0.1]	N	N
MountLondiani	0.572	0.010	[0.001, 0.1]	N	N
Narok	47.508	0.010	[0.001, 0.1]	N	Y
Samburu	95.455	0.010	[0.001, 0.1]	N	Y

Overall, in this experiment, Claude Sonnet 4 with a single zero-shot prompt is not currently suited for hyperparameterization when compared to the brute force grid search implemented in Ecoscope. There is limited current reasoning about the nonlinear dynamics of each site as well as model awareness of regime changes in the data. The LLM fails to represent the relationship between the ecological site factors and the smoothing penalty under cross-validation. For example, Marmanet has a visible drop in the data where the LLM interprets a low α, but evaluation determines a smooth curve can be used across the whole time range.

This represents some of the first applications of LLM-based hyperparameter search to domain-specific statistical models in the conservation field, and defines concrete gaps in current LLM context windows. A multi-shot prompt is worth exploring with labeled examples of validated α parameters as a way to teach the model nuanced selection. Applying Bayesian optimization as a preprocessing tool may also reduce experimental error.

6. Discussion

6.1. Predicting Regime Changes

GAMs reliably detect and represent abrupt regime changes that linear models cannot. MaraConservancies, Eburu, Loita, Marmanet, and MountLondiani all exhibit distinct deforestation phases separated by sudden transitions. OLS and GLM produce a single averaged rate that misrepresents both the stable and rapidly changing phases, while the GAM correctly fits each phase independently. A ranger using OLS at MaraConservancies in 2011 would have forecast gradual continued loss and missed the 35% collapse that followed from 2012 to 2016.

The GAMM can add a further benefit by learning shared change patterns across the nine sites. Areas undergoing similar transitions in the same period benefit from pooled information, improving estimates particularly where local data is sparse around the transition. This is a unique case of model improvement and cannot be applied to all nine sites evenly, which is why the per-site regression curves are also displayed to the researcher.

6.2. Use Cases for Ecological Prediction

The forest cover results demonstrate that the GAM/GAMM framework is well suited to generalized ecological time series due to its handling of nonlinear and unpredictable dynamics. The GAMM is particularly valuable in regional analyses where adjacent sites share ecological trends, as demonstrated by the five sites where cross-site pooling improved forecasting performance over the per-site GAM.

6.3. LLM vs Grid Search for Alpha Selection

The LLM initialization experiment shows that Claude Sonnet does not outperform random search for α selection on the nine forest cover sites. The model suggested $\alpha = 0.01$ for seven of nine sites regardless of input characteristics, correctly identifying low-α sites with strong regime changes but failing entirely on smooth high-α sites including Marmanet, Narok, and Samburu. The current grid search implementation remains the appropriate default. The gap in ecological intuition provides a concrete target for future work via few-shot prompting or Bayesian optimization.

7. Conclusion

This work presents a Python-based trend fitting framework for wildlife conservation practitioners, implemented within the Ecoscope library and evaluated on nine Kenyan forest monitoring sites spanning 2001–2024. The framework operationalizes four regression models — OLS, GLM, GAM, and GAMM — each relaxing one assumption from its predecessor. Pre-processing is carried out so that practitioners interact only with raw data and interpretable outputs.

The primary empirical finding is that the GAM or GAMM outperforms OLS on seven of nine sites for rolling origin forecasting, with the advantage largest at sites exhibiting abrupt regime changes. The GAMM with per-site random intercepts outperforms the per-site GAM on five of nine sites by leveraging shared trend structure across the monitoring network. Sites with highly individual dynamics retain a per-site GAM advantage. A secondary case study on a year of hourly elephant movement telemetry shows that the same pipeline transfers without modification to a far denser and noisier series, recovering a stable nonlinear trend where the linear model reports no change.

The LLM hyperparameter initialization experiment yields a negative result that is informative for future work. Under the prompting setup, Claude Sonnet 4 did not encode the relationship between ecological site characteristics and optimal smoothing parameters well. This defines a concrete research gap and motivates few-shot and Bayesian approaches as future directions.

Together these contributions address the growing need for widespread trend analysis tooling in ecological spaces — providing conservation organizations with statistical tools that are powerful enough to capture genuine environmental complexity, robust enough to perform reliably on sparse and irregular data, and accessible enough to deploy without requiring advanced statistical expertise. The implementation within Ecoscope ensures that the framework is available to the more than 800 EarthRanger sites currently operating worldwide.

8. Code

The full implementation is available across three public repositories:

- **Ecoscope library:** https://github.com/wildlife-dynamics/ecoscope
- **Trend fitting module:** ecoscope/analysis/trend_analysis.py
- **Forest cover workflow:** https://github.com/wildlife-dynamics/wt-hansen-deforestation
- **Elephant speed map workflow:** https://github.com/wildlife-dynamics/speedmap-trend

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